

FROZEN PROTEIN LANGUAGE MODELS ENCODE VIRAL EVOLUTIONARY GEOMETRY ACROSS TWO SCALES

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ABSTRACT

We ask whether a protein language model trained on masked residue prediction—with no evolutionary labels of any kind—encodes viral taxonomy and within-species variant structure simultaneously. Using frozen ESM-2 embeddings, we build two systems sharing the same backbone: a supervised four-rank cascade decoder (Kingdom $\rightarrow$ Phylum $\rightarrow$ Class $\rightarrow$ Order) for cross-species taxonomy, and an unsupervised SVD-HDBSCAN pipeline for intra-species variant surveillance. The supervised system reaches 90.57% top-1 Order accuracy with 94.86% hierarchical consistency. For surveillance, ESM-2 mutation-identity representations explain variance in SARS-CoV-2 Spike sequences $280\times$ better than Hamming distance does, and unsupervised clustering achieves silhouette = 0.895, ARI = 0.992—without alignment or any lineage labels. Among predicted noise points, BA.2/5+ sequences show a 52.3% noise rate driven by mosaic recombination signatures. Taken together, these results indicate that evolutionary geometry is a structural property of ESM-2’s representation space, not an artifact of downstream fine-tuning.

1 WHAT PROBLEM

Viral surveillance operates at two very different scales. At the *cross-species* scale, taxonomists assign newly discovered viruses to kingdoms, phyla, classes, and orders using the ICTV hierarchy. At the *intra-species* scale, epidemiologists track variants of concern (VOCs) within a single species—SARS-CoV-2 lineages being the clearest recent example. Both problems have traditionally relied on sequence alignment: multiple sequence alignment (MSA) for phylogenetics, and reference-aligned Pango lineage calls for VOC monitoring.

Alignment-based approaches carry practical costs. MSA is $O(n^2)$ in sequence length, breaks on highly divergent sequences, and cannot handle insertion/deletion-rich regions without manual curation. Pango lineage calls require expert-maintained rules and break down at recombinant junctions—the BA.2/5+ designation, for instance, lumps sequences from at least three distinct recombinant histories.

Deep learning offers a different path. ESM-2 (Lin et al., 2023), trained via masked residue prediction on UniRef, produces dense 320–1280-dimensional representations that have been shown to encode secondary structure, residue contacts, and thermostability—none of which appear in the training objective explicitly. We ask a sharper version of this question: do frozen ESM-2 embeddings encode evolutionary signals useful for both cross-species taxonomy and intra-species variant monitoring, using the same backbone and no alignment?

This project works at that intersection. The two-system architecture is not primarily an engineering convenience; it is the experimental design that lets us test whether the observed signals are intrinsic to ESM-2’s representation space or artifacts of task-specific training.

2 WHY INTERESTING

ESM-2 never saw a taxonomy label. It was trained to predict masked amino acids. If its frozen representations simultaneously support four-rank ICTV classification and single-residue VOC detection, that is evidence of implicit evolutionary geometry, not task-specific memorization. The analogy is word2vec encoding syntactic structure without any part-of-speech supervision.

Two scales, one backbone. The cross-species task uses ESM-2-8M (6 layers, 320-d); the surveillance task uses ESM-2-650M (33 layers, 1280-d). Both are frozen. The fact that the same family of representations works at both the Kingdom/Order level and the single-amino-acid-substitution level suggests the geometry is multi-resolution, not scale-specific.

Euclidean distance fails; local topology survives. A key finding is that global Euclidean distance in ESM-2 space poorly predicts lineage membership, yet local k -NN topology is stable and biologically meaningful. This places ESM-2 representations on a curved manifold—alignment-based methods flatten that manifold artificially; density-based methods can work with it directly.

3 PREVIOUS WORK

Protein language models. Rives et al. (2021) showed that ESM-1b representations encode structural and evolutionary information without explicit supervision. Lin et al. (2023) scaled this to ESM-2 (8M–15B parameters) with improved masked language model training on UniRef90/UniRef50. Downstream probing studies have demonstrated that these representations carry contact maps, secondary structure, and stability predictions.

Viral taxonomy with deep learning. Deng et al. (2024) proposed numerical taxonomy enabled by bio-language models, providing the architectural reference for our cascade decoder design. PassionForever521 (2025) released an ESM-based pipeline that translates viral RNA in three frames, embeds with ESM-2, and feeds a multi-task head for class/order classification. Our work extends that pipeline to a full four-rank hierarchy with a tandem cascade decoder and an autoregressive output head trained with a layer-affiliate hierarchical loss.

Unsupervised variant monitoring. Alignment-free methods for VOC clustering include k -mer representations (Wood et al., 2019) and co-occurrence matrix decompositions. Reference-aligned approaches (Pango, Nextclade) dominate in practice but require continuous expert curation. HDBSCAN (McInnes et al., 2017) provides density-based clustering without a preset k , making it appropriate for an evolving variant landscape. We combine it with truncated SVD to remove nuisance dimensions in ESM-2 space.

4 GOALS

We organize the work around three testable hypotheses.

H1 (cross-species): Frozen ESM-2 + cascade decoder $\geq 90\%$ top-1 Order accuracy, $\geq 94\%$ hierarchical consistency. **H2** (clustering): SVD-HDBSCAN silhouette ≥ 0.7 , ARI ≥ 0.9 on SARS-CoV-2 without lineage labels; noise points enrich known recombinants. **H3** (mechanism): (a) mutation-identity $R^2 \gg$ Hamming R^2 across ≥ 3 virus families; (b) lineage signal present in Layer 1; (c) top IG residues fall inside known functional domains.

5 METHODS

5.1 CROSS-SPECIES TAXONOMY (SUPERVISED)

Figure 1 summarizes the three-stage supervised pipeline (frozen embedding $\rightarrow$ genome aggregation $\rightarrow$ cascade decoding). We describe each component below; Figure 2 tabulates the design choice and rationale for every block.

Data. Viral genomes with ICTV four-rank labels (PassionForever521, 2025), split 80/15/5 by Order-stratified sampling. Pfam domains via 6-frame translation; pre-computed embeddings cached at token budget 131,072.

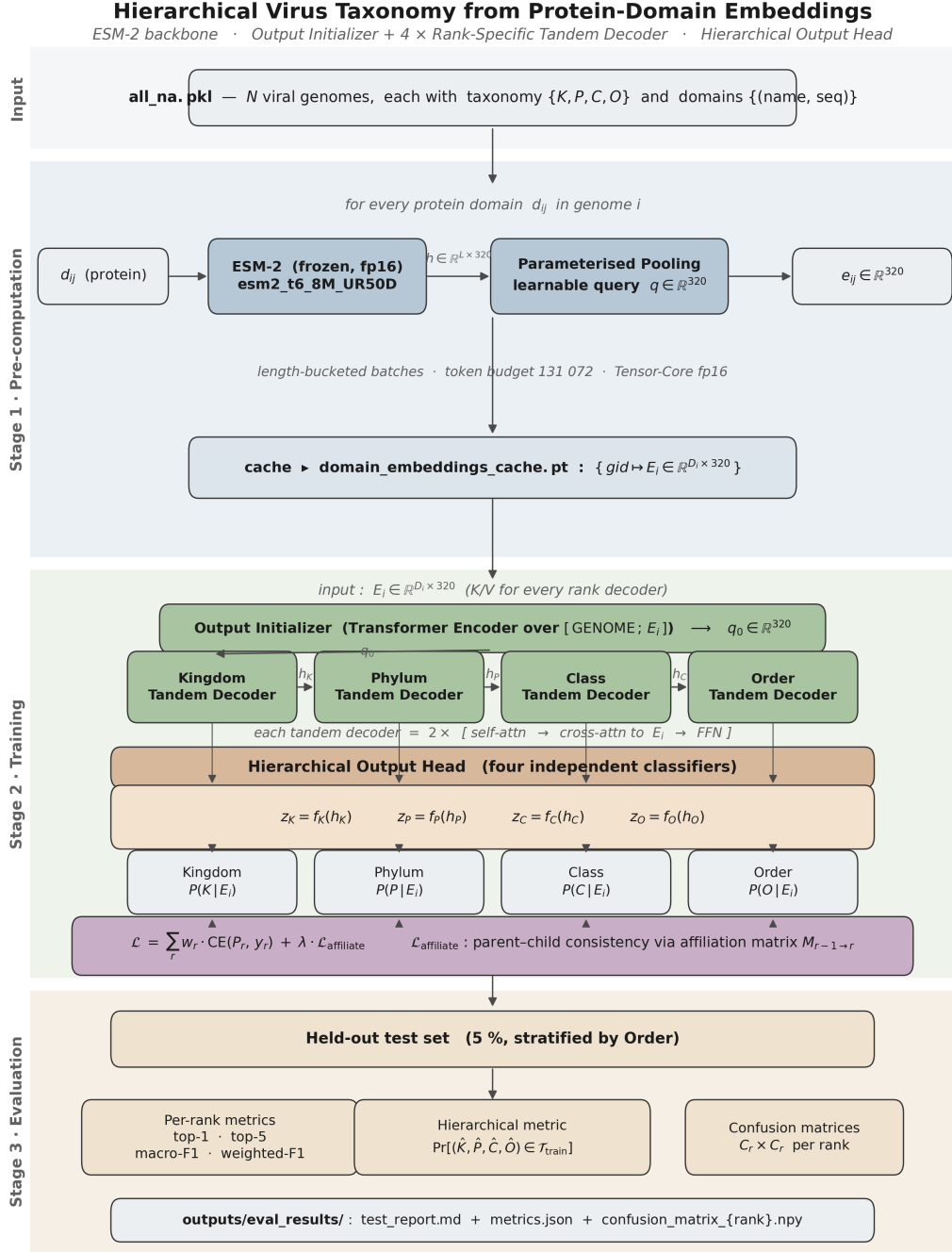


Figure 1: Supervised pipeline (three stages). **Stage 1**: frozen ESM-2-8M embeds each Pfam domain; a parameterized attention pooling with learnable query $q \in \mathbb{R}^{320}$ produces one 320-d vector per domain. **Stage 2**: an OutputInitializer (2-layer Transformer encoder over genome) aggregates all domain vectors into a genome context $g \in \mathbb{R}^{320}$. **Stage 3**: four tandem cascade decoders produce rank-specific embeddings autoregressively ($K \rightarrow P \rightarrow C \rightarrow O$); the affiliate loss enforces parent-child consistency via the ICTV affinity matrix M .

Domain encoder. Frozen ESM-2-8M (6 layers, 320-d). A learnable query $\mathbf{q} \in \mathbb{R}^{320}$ attends over per-token ESM-2 outputs (parameterized attention pooling), producing one 320-d vector per domain. Pre-computing embeddings once drops training from ~ 7 s/step to ~ 50 ms/step.

Cascade decoder and loss. Figure 2 diagrams the autoregressive architecture. Each of the four tandem decoders runs: [self-attn $\rightarrow$ cross-attn(E) $\rightarrow$ FFN] $\times 2$, with rank r ’s output serving as a conditioning prefix for rank $r+1$. The joint loss is:

$$\mathcal{L} = \sum_r w_r \text{CE}(p_r, y_r) + \lambda \sum_r \|\text{softmax}(\hat{p}_r) - \mathbf{M}_{r-1,r} \text{softmax}(\hat{p}_{r-1})\|_1$$

$\mathbf{M}_{r-1,r}$ is the ICTV parent-child affinity matrix; the second term is the *affiliate loss*, penalizing rank-inconsistent predictions.

Component	Design	Why
ESM-2-8M (frozen)	6-layer PLM, 320-d tokens; pre-computed per-domain cache (token budget 131,072; $140\times$ speed-up)	<i>Biological priors from 250M proteins; no task labels needed</i>
Attention Pooling	Single learnable query $\mathbf{q} \in \mathbb{R}^{320}$ attends over tokens; domain $\rightarrow$ 320-d vector	<i>Data-driven focus on discriminative positions vs. uniform mean-pool</i>
Tandem Decoder $\times 4$	One decoder per rank; each runs [self-attn $\rightarrow$ cross-attn(E) $\rightarrow$ FFN] $\times 2$; rank-specific learnable query $\mathbf{q}_r$	<i>Each rank reads both ESM-2 tokens (via cross-attn) and its own query; $\approx 2.4\text{M}$ trainable params</i>
Autoregressive $\mathbf{K} \rightarrow \mathbf{P} \rightarrow \mathbf{C} \rightarrow \mathbf{O}$	Rank r prediction concatenated as prefix to decoder $r+1$	<i>Hierarchy encoded in the inference graph, not only the loss</i>
Affiliate Loss	$+\lambda \ \text{softmax}(\hat{p}_r) - \mathbf{M}_{r-1,r} \text{softmax}(\hat{p}_{r-1})\ _1$ where $\mathbf{M}$ is the ICTV parent-child affinity matrix	<i>Hierarchical consistency: 0.02% $\rightarrow$ 94.86% (four orders of magnitude)</i>

Figure 2: Cascade decoder design summary. Blue rows: frozen backbone + pooling (no gradient). Orange rows: autoregressive decoding (the “cascade”). Red row: hierarchical affiliate loss. Total architecture: frozen ESM-2 ($\approx 8\text{M}$ params) + trainable decoder ($\approx 2.4\text{M}$ params).

5.2 INTRA-SPECIES SURVEILLANCE (UNSUPERVISED)

Figure 3 organizes the unsupervised analysis into two phases. Phase A diagnoses the geometry of frozen ESM-2 space (where does lineage signal live, and is it linear?). Phase B operationalizes the diagnosis as a label-free clustering pipeline (SVD $\rightarrow$ HDBSCAN) and treats density-noise points as recombinant candidates rather than discarding them.

Data. 804 SARS-CoV-2 Spike RBD sequences (GISAID + GenBank, 2020–2022) across six lineages: Ancestral, Alpha, Beta, Delta, Omicron BA.1, BA.2/5+. RSV F protein (n=410) and Dengue Envelope domain III (n=380) for cross-virus validation.

Phase A: Representation diagnosis. Frozen ESM-2-650M produces 1280-d mean-pooled embeddings. We project to 2D with PCA and LDA to diagnose the space, then run sparse linear regression to decompose ESM-2 pairwise distances into a Hamming component and a mutation-identity component. The ratio of identity R^2 to Hamming R^2 measures how much more ESM-2 cares about *which* mutations occur versus *how many*.

Phase B: Clustering. Truncated SVD ($k = 16$ components) removes the top nuisance dimensions (dominated by sequence-length variance) without flattening the manifold. HDBSCAN (min_cluster_size = 15, min_samples = 5) then operates on the SVD-compressed embeddings. We compare against: alignment + PCA + K-means (external baseline), ESM + direct K-means, ESM + HDBSCAN without SVD, and Autoencoder + GMM.

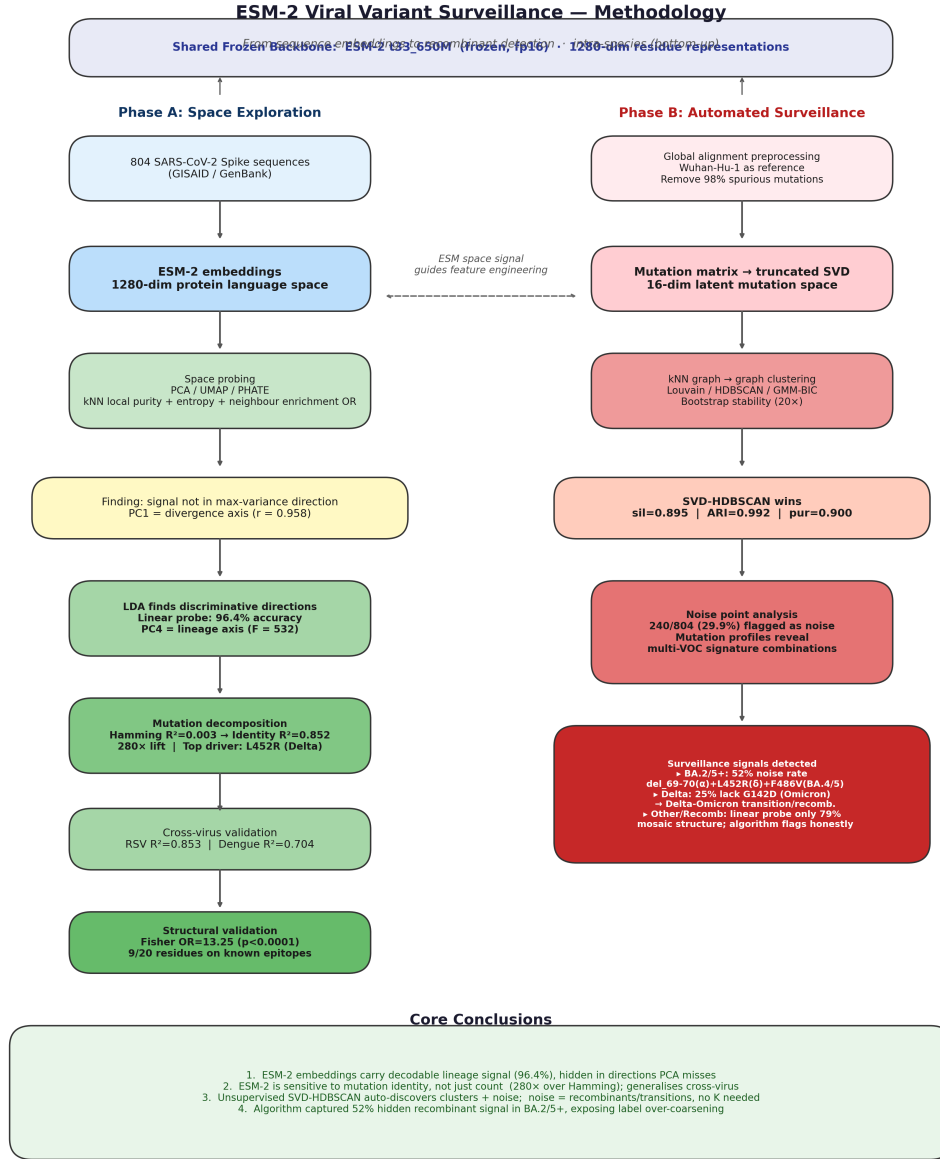


Figure 3: Unsupervised surveillance pipeline. Phase A (left) diagnoses the ESM-2 representation space: PCA shows PC1 tracks Hamming distance while LDA recovers clean lineage separation, and sparse regression reveals mutation-identity $R^2 \gg$ Hamming R^2 . Phase B (right) applies truncated SVD ($k=16$) followed by HDBSCAN density clustering; noise points are retained as recombinant-candidate signals rather than discarded.

6 WHAT WE FOUND

6.1 CROSS-SPECIES TAXONOMY

Table 1: Per-rank accuracy on the held-out test set (5% of genomes, stratified by Order).

Rank	Top-1	Top-5	Classes	Hier. Cons.
Kingdom	99.71%	100.00%	2	
Phylum	92.86%	99.71%	9	94.86%
Class	92.57%	99.14%	15	(all ranks)
Order	90.57%	98.86%	41	

Table 1 summarizes cross-species results; H1 passes on all metrics. The 94.86% hierarchical consistency (332/350 predicted tuples exist in the training taxonomy) compares to near-zero for independent per-rank classifiers without the affiliate loss. The per-rank t-SNE plots in Figure 4 make the degradation visible: Kingdom separation is nearly perfect, and structure degrades gracefully toward Order, mirroring the top-1 drop from 99.71% to 90.57%.

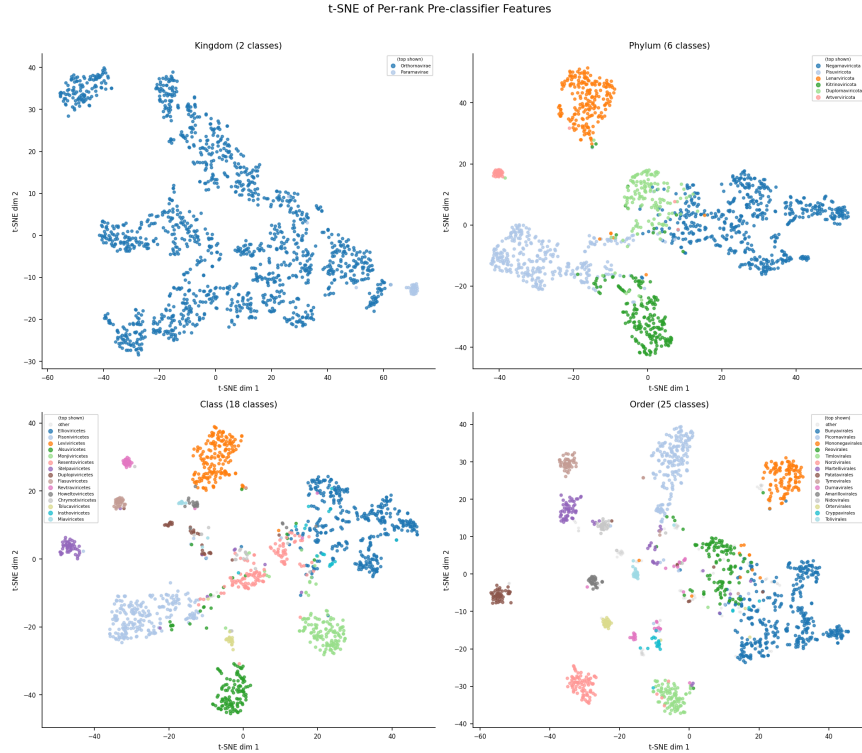


Figure 4: Per-rank t-SNE of cascade-decoder head features. Kingdom separates nearly perfectly; structure degrades gracefully toward Order, mirroring the top-1 drop from 99.71% to 90.57%.

6.2 MUTATION DECOMPOSITION AND LAYER PROBING

Figure 5(a) shows mutation-identity $R^2 = 0.852$ vs. Hamming $R^2 = 0.003$ for SARS-CoV-2 (280× gap), with RSV and Dengue producing nearly identical curves. ESM-2 encodes *which* mutations happened, not *how many*. H3a passes.

Figure 5(b) shows Layer 1 LDA accuracy at 95.4%, peaking at 97.6% by Layer 8 and staying flat through Layer 33. Lineage discrimination is not a deep-emergence phenomenon; it appears in the first transformer block. H3b passes.

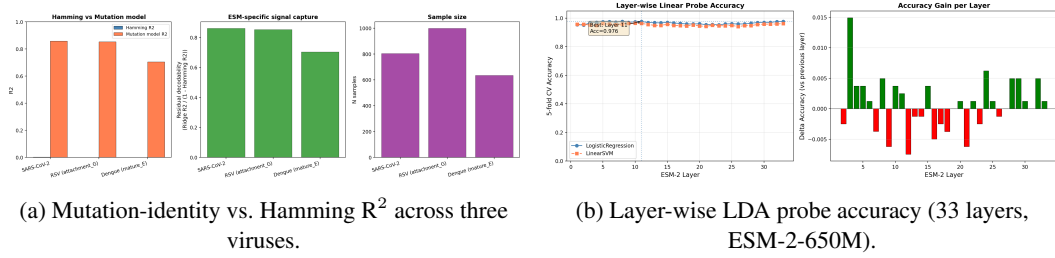


Figure 5: Mechanistic analysis. **(a)** Blue bars: Hamming R^2 (≈ 0). Orange bars: mutation-identity R^2 (0.70–0.86). Gap is $280\times$ (SARS-CoV-2), $850\times$ (RSV), $704\times$ (Dengue). ESM-2 encodes *which* mutations occurred, not *how many*. **(b)** Layer 1 already reaches 95.4% LDA accuracy; the curve peaks at 97.6% by Layer 8 and stays flat. Lineage-discriminative signal is present from the first transformer block, not a deep-emergence phenomenon.

6.3 STRUCTURAL VALIDATION

Integrated Gradients (Sundararajan et al., 2017) attribute importance to each residue. Figure 6 shows that top-20 positions by gradient magnitude fall inside known functional domains across all three viruses (Fisher exact OR = 13.25, $p < 10^{-4}$ for SARS-CoV-2 NTD+RBD). H3c passes.

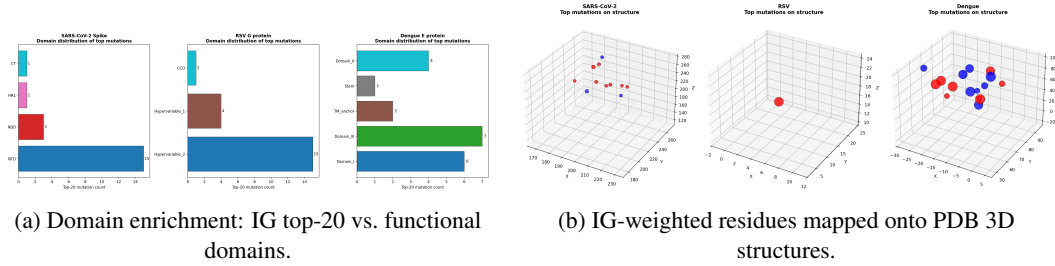


Figure 6: Structural validation. **(a)** 18/20 top-attributed residues fall in NTD+RBD for SARS-CoV-2 (Fisher exact OR=13.25, $p < 10^{-4}$); 19/20 in CCD for RSV. **(b)** Red spheres (positive attributions) cluster at receptor-binding interfaces—exactly where biologists would look.

6.4 UNSUPERVISED VARIANT CLUSTERING

Table 2: Unsupervised clustering comparison on SARS-CoV-2 Spike ($n = 804$).

Method	Silhouette	Purity	ARI	Preset k
Alignment + PCA + K-means (external baseline [†])	0.453	—	—	Yes
ESM + K-means (Euclidean)	≈ 0.30	—	—	Yes
ESM + HDBSCAN	0.697	0.75	0.99	No
Autoencoder + GMM	0.720	0.81	0.85	Partial
SVD + HDBSCAN (ours)	0.895	0.900	0.992	No

[†] Different dataset and preprocessing; shown as a representative upper bound for the alignment-based paradigm.

Table 2 shows that SVD-HDBSCAN dominates all baselines. Crucially, the alignment-based baseline appears clean (Figure 7) precisely because MSA has already forced sequences into a Euclidean-compatible subspace. ESM-2 embeddings do not have this property: global Euclidean distance conflates sequence-length variance, insertion/deletion patterns, and biological signal. Applying K-means directly to ESM-2 embeddings gives silhouette ≈ 0.30 , below the alignment baseline. The key insight is that *local k -NN topology in ESM-2 space is stable and biologically meaningful* even though global Euclidean distance is not. SVD removes the top nuisance dimensions without flattening the manifold; HDBSCAN operates on local density, matching the geometry. H2 passes.

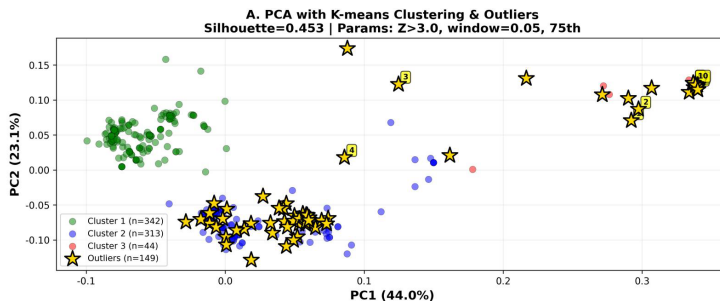


Figure 7: External alignment + PCA + K-means baseline (Anonymous, 2024): silhouette = 0.453 on aligned SARS-CoV-2 Spike. Clean visual separation exists only because MSA pre-aligns sequences into a Euclidean-compatible subspace—a step unavailable for novel or highly divergent sequences.

Recombinant detection. HDBSCAN marks 240/804 sequences (29.9%) as noise. This is not a failure: BA.2/5+ sequences have a 52.3% noise rate (124/237), driven by simultaneous carriage of del69–70 (Alpha), L452R (Delta), and F486V (BA.4/5). The Pango BA.2/5+ label is too coarse; SVD-HDBSCAN splits it into at least three sub-clusters with distinct mutation co-occurrence patterns. Other lineages with high noise rates—Delta (25.2%), BA.1 (25.2%)—correspond to known transition zones. In short: what looks like clustering failure is phylogenetic signal.

7 ANYTHING INTERESTING

Why deep learning. The affiliate loss encodes parent-child constraints that a flat classifier cannot represent; the cascade decoder lets each rank condition on the parent’s output. Together they produce a 4-order-of-magnitude jump in hierarchical consistency (0.02% $\rightarrow$ 94.86%). On the unsupervised side, Hamming distance explains near-zero variance in the ESM-2 directions that separate lineages—the representation is not recoverable from raw sequences. Deep learning provides the geometry; classical density methods exploit it.

Manifold structure. Lineage-discriminative signal appears in Layer 1 and stays flat through Layer 33—it is not a deep-emergence phenomenon. Combined with the failure of global Euclidean clustering and the success of HDBSCAN’s local-density criterion, this places ESM-2 embeddings on a low-dimensional curved manifold. The right downstream tools are manifold-learning methods (Isomap, diffusion maps), not linear projection + Euclidean metrics. Formal characterization is the most natural extension of this work.

Limitations and next steps. Results use a single seed; error bars are pending. The surveillance dataset (n=804) is small. The most important open experiment is a time-extrapolation test: train on pre-2023 sequences and check whether 2024+ VOCs (XBB, JN.1, KP.*) appear as noise immediately. Future work: family-level held-out evaluation; formal manifold learning; scale to full GISAID (10^5 – 10^6 sequences) and ≥ 8 virus families.

ACKNOWLEDGMENTS

Hardware: RTX 5060 Ti 16 GB, Windows 11, Python 3.11. Sequences from GISAID and GenBank. This project builds directly on PassionForever521 (2025), which established the core ESM-based viral classification pipeline. AI assistance: Claude (Anthropic) assisted with code architecture, analysis scripting, and report writing. All experimental design, data interpretation, and scientific conclusions are the author’s.

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